EXPERIMENT : 19

MATLAB — Introductory Simulation of a Selected 6G Technology

OBJECTIVEES :

• To select and study a manageable 6G technology such as THz communication, Intelligent Reflecting Surface (IRS), or Integrated Sensing and Communication (ISAC).

• To implement a basic MATLAB model or parameter study for the selected 6G technology.

• To generate and analyze at least one relevant performance metric such as path loss, received power, coverage gain, achievable rate, or sensing accuracy.

• To understand the relevance of the selected technology to 6G networks, non-terrestrial networks (NTN), and emerging wireless standardization activities.

PROGRAME CODE :

```
clc;
clear;
close all;

%% 1. 6G IRS-Assisted Communication Parameters
c = 3e8;                % Speed of light (m/s)
f = 100e9;              % Carrier frequency (100 GHz)

Pt = 20;                % Transmit power (dBm)
Gt = 10;                % Transmitter antenna gain (dBi)
Gr = 10;                % Receiver antenna gain (dBi)

d1 = 50;                % Transmitter-to-IRS distance (m)
d2 = 50;                % IRS-to-receiver distance (m)

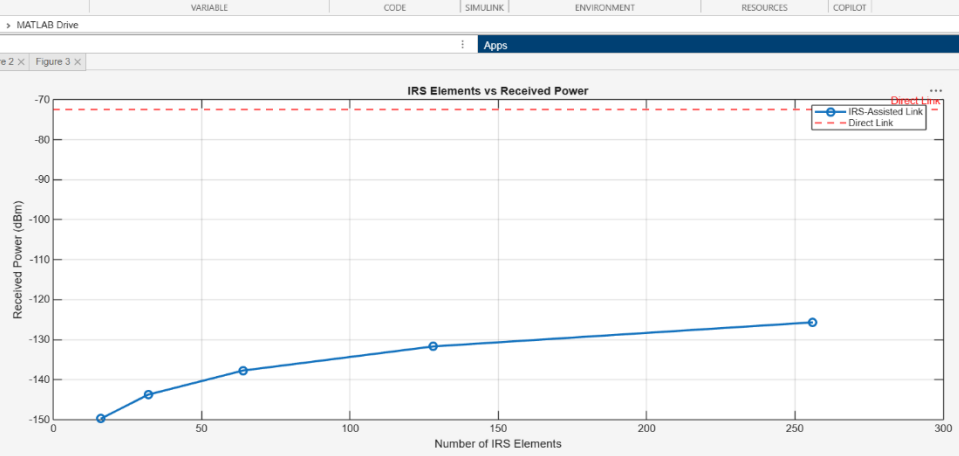
eta = 0.8;              % IRS reflection efficiency

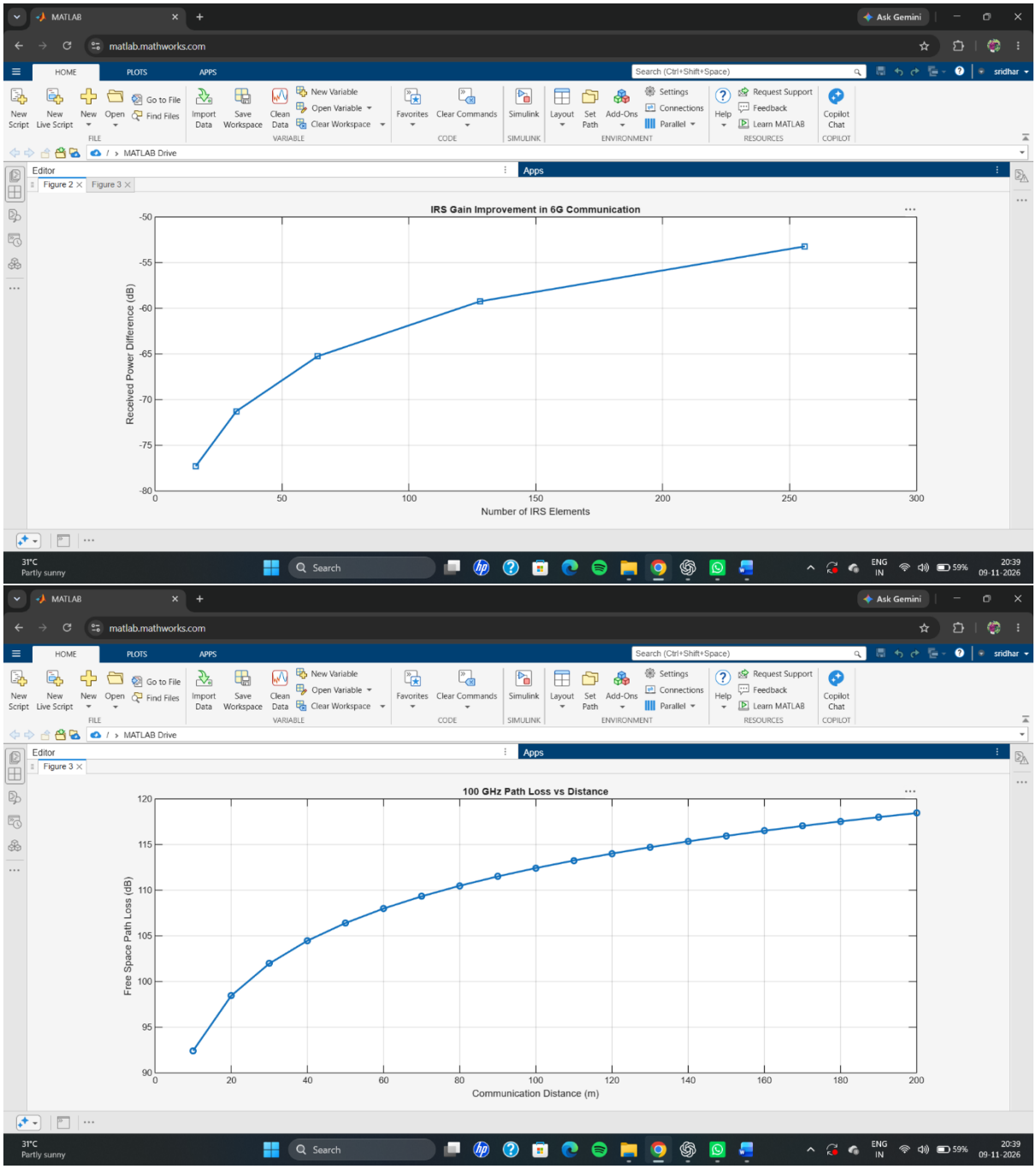
N = [16 32 64 128 256];  % Number of IRS elements

%% 2. Calculate Wavelength
lambda = c/f;

%% 3. Direct Communication Link
d_direct = d1 + d2;

% Free Space Path Loss for direct link
PL_direct = 20*log10(4*pi*d_direct/lambda);
```





RESULT : The MATLAB simulation of 6G Intelligent Reflecting Surface (IRS)-Assisted Communication was successfully implemented at a carrier frequency of 100 GHz. The performance of direct and IRS-assisted communication links was analyzed for different numbers of reflecting elements (16, 32, 64, 128, and 256). The results demonstrated that increasing the number of IRS elements improves the received signal power under the assumed coherent reflection model. The simulation also illustrated the relationship between communication distance and free-space path loss. This study demonstrates the potential of IRS technology to enhance signal coverage and support reliable communication in future 6G wireless networks.